

Connected & Aware — Intelligent IoT Ecosystems for Real-World Data

Theme: Develop intelligent IoT ecosystems that connect physical devices with software platforms to collect, process, visualize, and act on real-time data — combining sensors, microcontrollers, communication protocols, edge computing, and cloud/software systems to solve practical problems.

DOMAIN

IoT & Edge Systems

CATEGORY

HARDWARE

SOFTWARE

PS TYPE

Open Innovation

BACKGROUND

Physical spaces — homes, farms, factories, vehicles, and cities — generate a constant stream of real-world signals: temperature, motion, location, load, air quality, and more. Most of this data is either never captured or captured and left unused, because building a reliable pipeline from a sensor to a decision is hard: it means handling unreliable connectivity, noisy readings, limited device power, and turning raw signals into something a person or a system can actually act on in time to matter.

PROBLEM STATEMENT

Design and build an IoT-based system that senses a real-world condition through connected hardware, transmits that data reliably to a software platform, and processes it — at the edge, in the cloud, or both — to visualize what is happening and trigger a meaningful automated action or alert, closing the loop between the physical and digital world.

EXPECTED FOCUS AREAS

- Sensor and microcontroller-based data acquisition from the physical environment
 - Edge processing to reduce latency, bandwidth use, and dependence on connectivity
 - Automated actions or alerts triggered by defined thresholds or learned patterns
- Reliable communication protocols (Wi-Fi, BLE, LoRa, MQTT) between devices and the cloud
 - Real-time dashboards and visualization of live sensor data
 - Power efficiency and robustness for devices operating unattended in the field

EXPECTED OUTCOME

A working IoT prototype — connected hardware plus a software platform — that continuously captures real-world data, presents it through a live dashboard, and demonstrably acts on that data (via an alert, automated control, or triggered workflow) without needing constant manual monitoring.

Solutions will be evaluated on end-to-end reliability of the sensor-to-action pipeline, quality of data visualization, sensible use of edge vs. cloud processing, and how directly the system solves a practical, real-world problem.

IMPACT

REAL-TIME AWARENESS

Turns previously invisible physical conditions into visible, actionable data.

AUTOMATION

Reduces manual monitoring by letting systems sense, decide, and act on their own.

CROSS-DOMAIN REACH

Applicable across agriculture, healthcare, industry, homes, and smart-city settings.